



CSE350: Digital Electronics and Pulse Techniques

Lab Guide

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Exp-01: Flash Analog to Digital converter (ADC)

General Instructions:

1. Make two power rails of 5V and -5V from the **variable dc power supply** (top left) of the trainer board. Connect the saturation voltages (VH and VL) of the op amps with the power rails. You might need to adjust the knob to get the desired voltage.
2. We will use the built-in LEDs (top right) of the trainer board.
3. Remember, we are using an active low encoder. If the binary output is 01, the LED connected with the LSB bit will be OFF, and the LED connected with the MSB bit will be ON.
4. During measurements of the output voltage, make sure you have **opened the output terminal first** and then take measurements. If you take measurements while the led is connected, you will get the voltage across the led. As a result, you won't get the actual output voltage.

Exp-02: Analysis of the binary weighted and R-2R D/A converter

The lab sheet for Experiment-02 is very self-explanatory. Still, there are some general instructions we would like to remind you of.

General Instructions:

1. Connect the saturation voltages (V_H and V_L) of the op amps with the **variable dc power supply** of the trainer board. Set the saturation voltages to 8V and -8V. You might need to adjust the knob to get the desired voltage.
2. **Don't connect the input pins (A, B, and C) with the switches of the trainer board**, as there is a current limit to the switches.
3. Make a power rail of 5V from the trainer board or the DC power supply. **If you wish to use the DC power supply, don't forget to connect the grounds of the trainer board and the DC power supply.**
4. If you need to turn an input low, don't leave that open. **Connect that pin with the ground.**
5. **Don't forget to add labels and titles to the graph.**

Exp-03: Analysis of Triangular Wave Generator

General Instructions:

1. Connect the saturation voltages (VH and VL) of the op amps with the **variable dc power supply** of the trainer board. Set the saturation voltages to 8V and -8V instead of 15V and -15V for safety reasons.
2. Press the Measure button to check the peak-to-peak, maximum voltage, minimum voltage, frequency, etc.



3. If the floating display shows “OFF” for the parameters of the voltage, press the following buttons.
4. Take the pictures of the waveforms. You need to add them at the end of the report. **Don't forget to add labels and titles.**
5. You need to use the **cursor button** (beside the Measure button) to find the rising and falling time.

Exp-04: Implementing Diode Logic (DL), Resistor Transistor Logic (RTL) and Diode Transistor Logic (DTL) gates

General Instructions:

1. For all tasks, connect the input pins to the trainer board switches and connect V_{CC} to the trainer board's 5V supply.
2. You can also connect V_{CC} and input pins with a DC voltage source, but as we limit the current through a DC voltage source, the output LED may appear very dim. We can solve this problem by increasing the current limit. But we don't want to change the current limit for safety reasons. That's why it is suggested to use the trainer board.
3. **When measuring the output voltage, ensure that the LED is disconnected.** Otherwise, the measured voltage will include the voltage drop across the LED.
4. If both the trainer board and the DC voltage source are used, make sure their grounds are connected.

Lab Task - 01:

1. Set one input of the DTL NAND gate to 5V (logic HIGH) using the trainer board or a DC voltage source.
2. Connect the other input to either a DC voltage source or the variable voltage knob of the trainer board. Gradually increase the input voltage and observe the output LED.
3. When the LED turns OFF, record the corresponding input voltage.

Exp-05: Implementing NMOS logic gates

General Instructions:

1. For all tasks, use the DC voltage source to supply V_{DD} and set the current limit to **0.3 mA**.
2. Do **not** take V_{DD} from the trainer board, as it does not provide current limiting. Without current limiting, the MOSFETs may overheat, leading to incorrect output voltages and power measurements.
3. The gate voltage of MOSFETs can be supplied either from the DC voltage source (by creating a power rail) or from the trainer board switches. This is acceptable because the MOSFET gate draws negligible DC current.
4. If both the DC voltage source and the trainer board are used, ensure that their **grounds are connected together**. Otherwise, the circuit will not operate correctly.
5. **A common mistake in total power calculation:** always calculate power using

$$P = V_{DD} \times I_{DD}$$

Do **not** use the gate voltage in this calculation, as it may differ from V_{DD} and lead to incorrect results.

Task 2:

1. When the input is low, the output voltage should be approximately $V_{DD} - V_{tn}$. Since the output is not exactly V_{DD} , the NMOS transistor is said to pass a “**weak 1**.” This is your task to find out why this happens.
2. When you are going to measure the current using a multimeter, ensure that the **red** knob of the probe is placed on the 10A max input of the multimeter. After that put the red wire where it was before.